INS 3121
SOIL MECHANICS

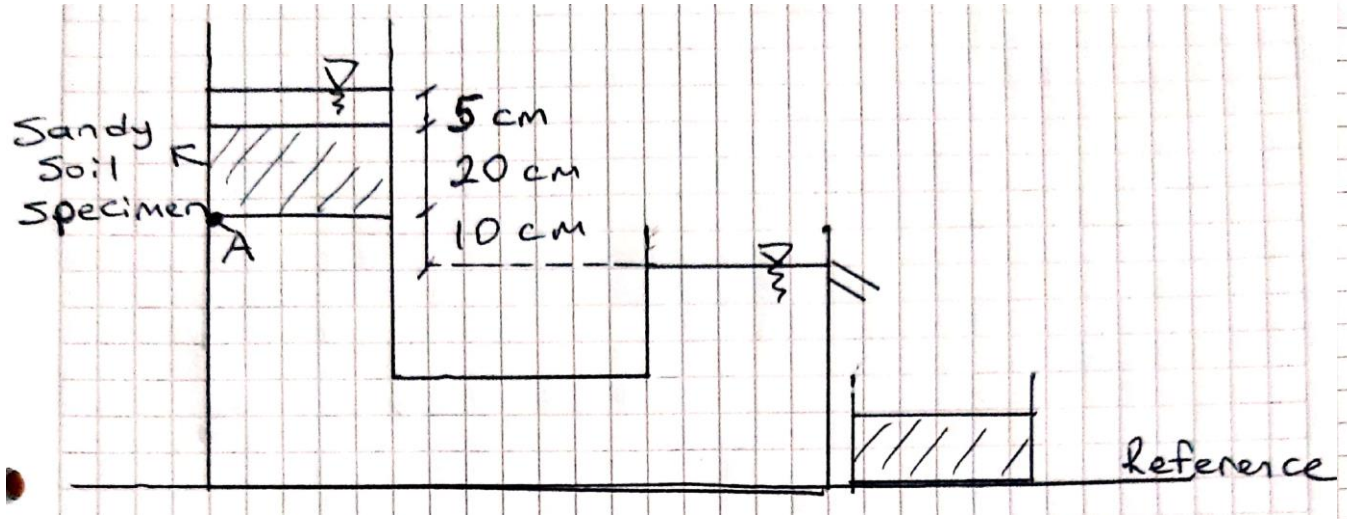
Examples:
Permeability and Seepage

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Example 1



Area of cross section $\Rightarrow A = 30\ cm^2$

$m_s = 1100g$ (Dry soil mass)

$G_s = 2.65$ (specific gravity)

Water collected in a period of 100 sec is $250\ cm^3$

($t = 100$ sec $Q = 250\ cm^3$)

A constant head test, shown in figure was performed to determine hydrolic conductivity of sandy soil

a) Determine hydraulic conductivity of the soil (k)

b) Determine discharge velocity (V) and seepage velocity (V_s)

Solution 1

Head difference is equal to water level difference

$\Delta h = 35$ cm

a) $k = \frac{Q \cdot L}{\Delta h \cdot A \cdot t} = \frac{250 \cdot 20}{35 \cdot 30 \cdot 100} = 4{,}76 \times 10^{-2}$ cm/sec.

Q : Volume of water collected

L : Length of soil specimen

Δh : head difference

A : Cross section area of soil specimen

t : duration of water collection

Solution 1

b) $V = k \cdot i$

$V = k \cdot \frac{\Delta h}{L} \Rightarrow 4,76 \times 10^{-2} \times \frac{35}{20} = 8.33 \times 10^{-2}$ cm/sec.

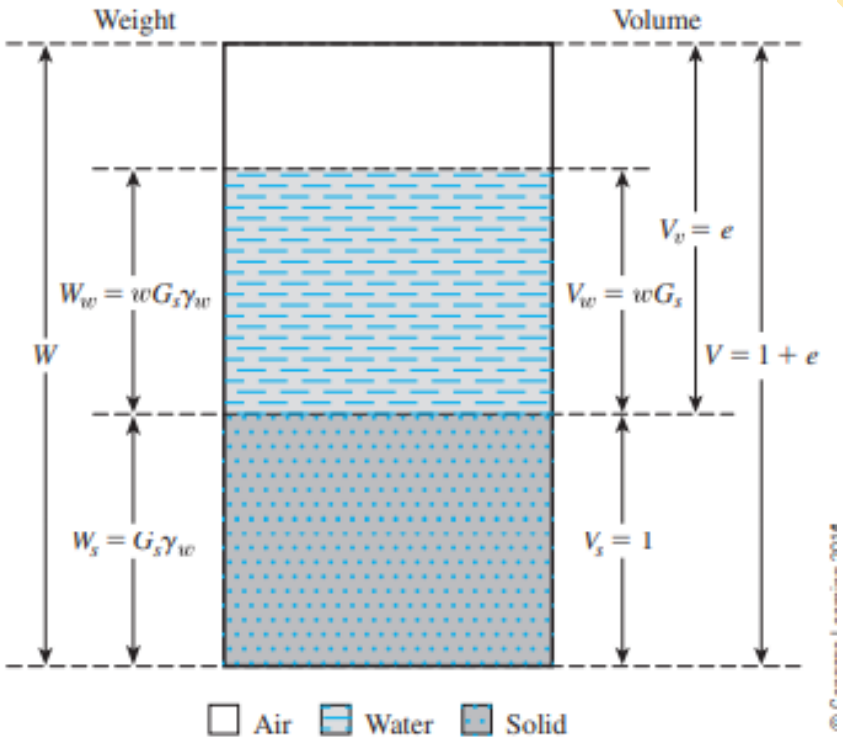
V: discharge velocity. We take all area in to consideration

$V_s = V \cdot \left(\frac{1+e}{e}\right) \Rightarrow V \cdot \frac{1}{n} \Rightarrow e = ?\quad n = ?$

$\rho_{dry} = \frac{m_s}{V} \Rightarrow \frac{1100}{30 \times 20} = 1,83\ g/cm^3$

$\rho_{dry} = \frac{G_s \cdot \rho_w}{1+e} \Rightarrow 1,83 = \frac{2,65 \times 1}{1+e} \Rightarrow e = 0,45$

$V_s = V \cdot \left(\frac{1+e}{e}\right) \Rightarrow 8,33 \times 10^{-2} \left(\frac{1+0,45}{0,45}\right) = 28,16 \times 10^{-2}$ cm/sec



Example 1 (Optional)

A sand sample of 35 cm^2 cross sectional area and 20 cm long was tested with a constant head permeameter.

Head difference (Δh): 60 cm

Discharge (Q): 120 mL

Duration (t): 6 min.

Dry mass of sand used for the test: 1120 g

- Specific Gravity (G_S): 2.68

Determine,

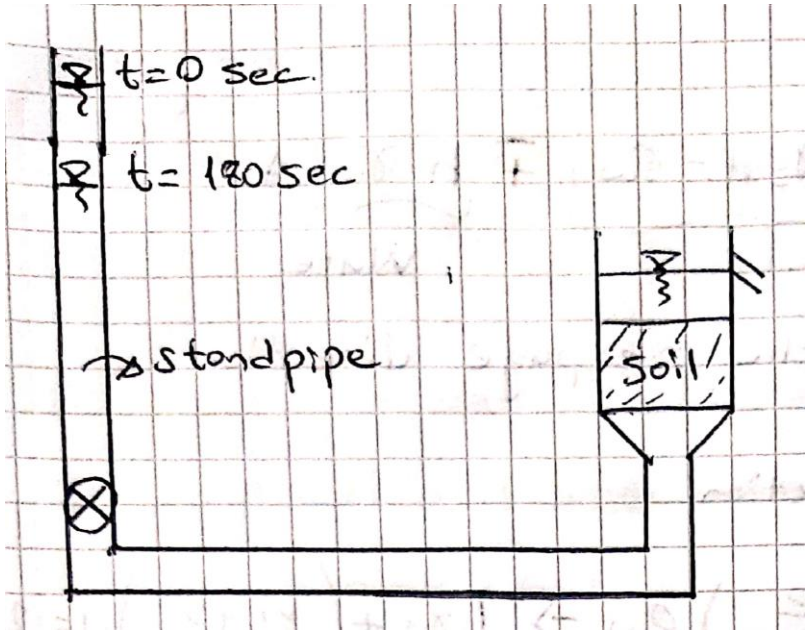
a) Hydraulic Conductivity in cm/sec

b) Discharge velocity and seepage velocity

a) $k = 3.175 \times 10^{-3}$ cm/sec

b) $V = 9.52 \times 10^{-3}$ cm/sec, $V_s = 2.36 \times 10^{-2}$ cm/sec.

Example 2



For a falling-head permeability test, the following values are given:

- Length of specimen = 200 mm.
- Area of soil specimen = 1000 mm^2
- Area of standpipe = 40 mm^2
- Head difference at time, t=0 = 500 mm
- Head difference at time, t = 180 sec = 300 mm

Determine the hydraulic conductivity of the soil in cm/sec.

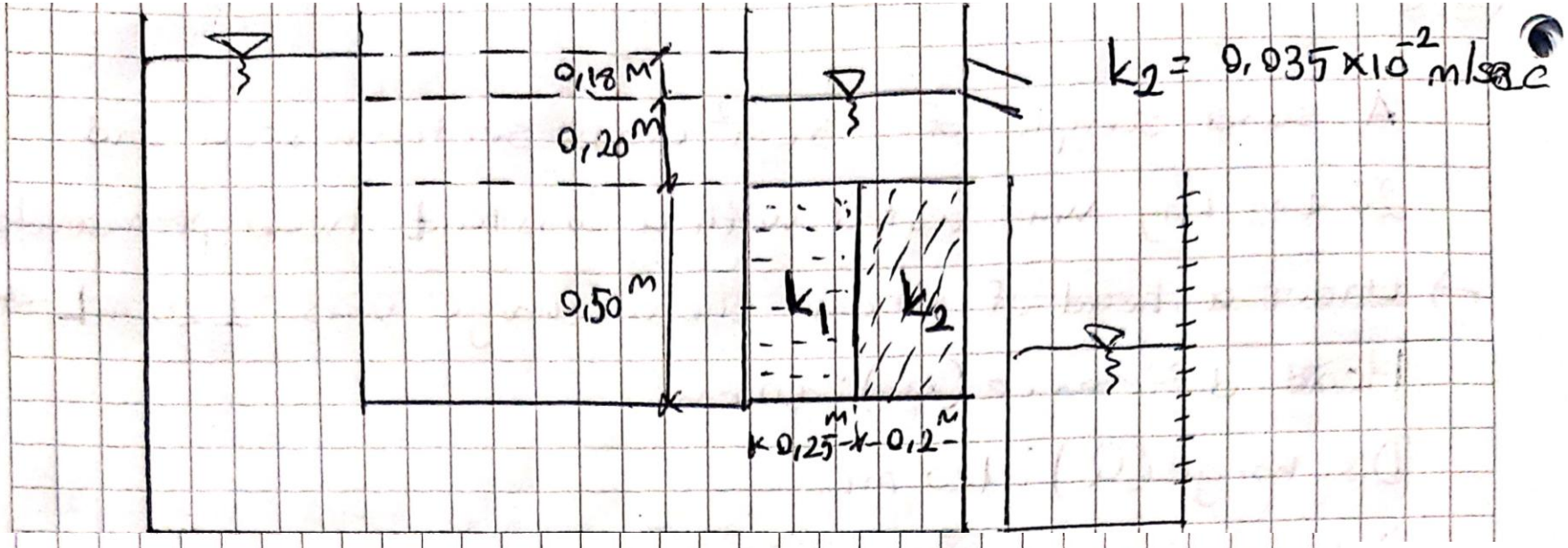
$$k = 2.303 \frac{aL}{At} \log_{10} \frac{h_1}{h_2}$$

$$k = 2.303 \frac{a \cdot L}{A \cdot t} \log_{10} \left(\frac{h_1}{h_2} \right)$$

$$k = 2.303 \frac{(40)(200)}{(1000)(180)} \log_{10} \left(\frac{500}{300} \right)$$

$$= 2.27 \times 10^{-2} \text{ cm/sec}$$

Example 3



In the figure two soil specimen with different k value are shown with constant head permeability test equipment. Calculate the permeability coefficient of 1st soil (k_1) if collected water amount is equal to 648 mL in 50 sec.

(The length of test equipment perpendicular to the figure plane is 0,3 m)

Solution 3

$q = k \cdot i \cdot A$, $Q = q \cdot t \Rightarrow q = \frac{648 \times 10^{-6}}{50} = 1.296 \times 10^{-5}\ m^3/sec$

$q = q_1 + q_2$, $A_1 = \frac{\pi \cdot 0.25^2}{4} = 0.049\ m^2$, $A_2 = \frac{\pi \cdot 0.2^2}{4} = 0.031\ m^2$

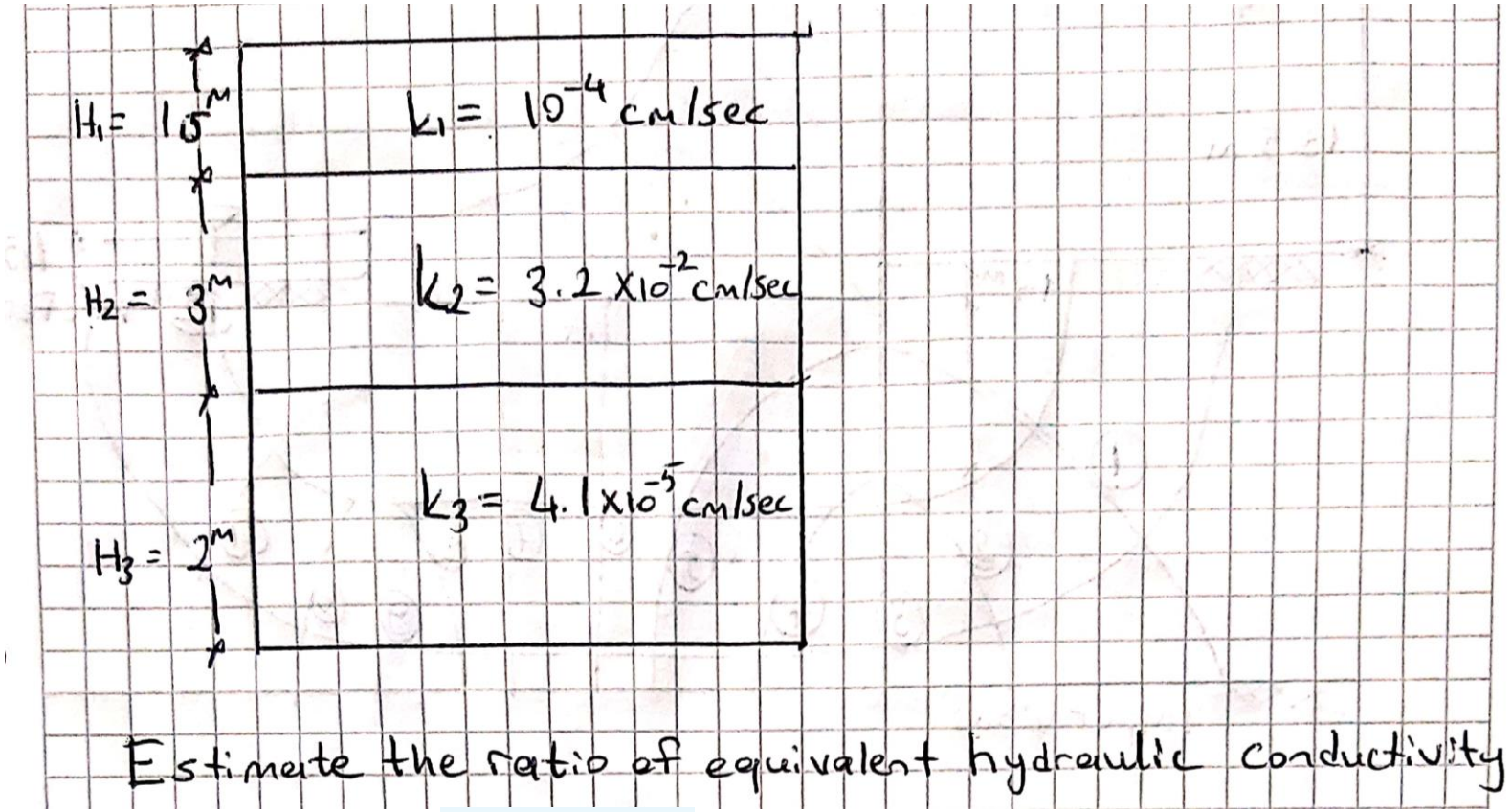
$q_1 = k_1 \cdot A_1 \cdot i$, $q_2 = k_2 \cdot A_2 \cdot i$

$i = \frac{\Delta h}{L} = \frac{0.18}{0.5} = 0.36$

$1.296 \times 10^{-5} = k_1 \times 0.049 \times 0.36 + 9.035 \times 10^{-2} \times 0.031 \times 0.36$

$k_1 = 5.133 \times 10^{-4}\ m/sec$

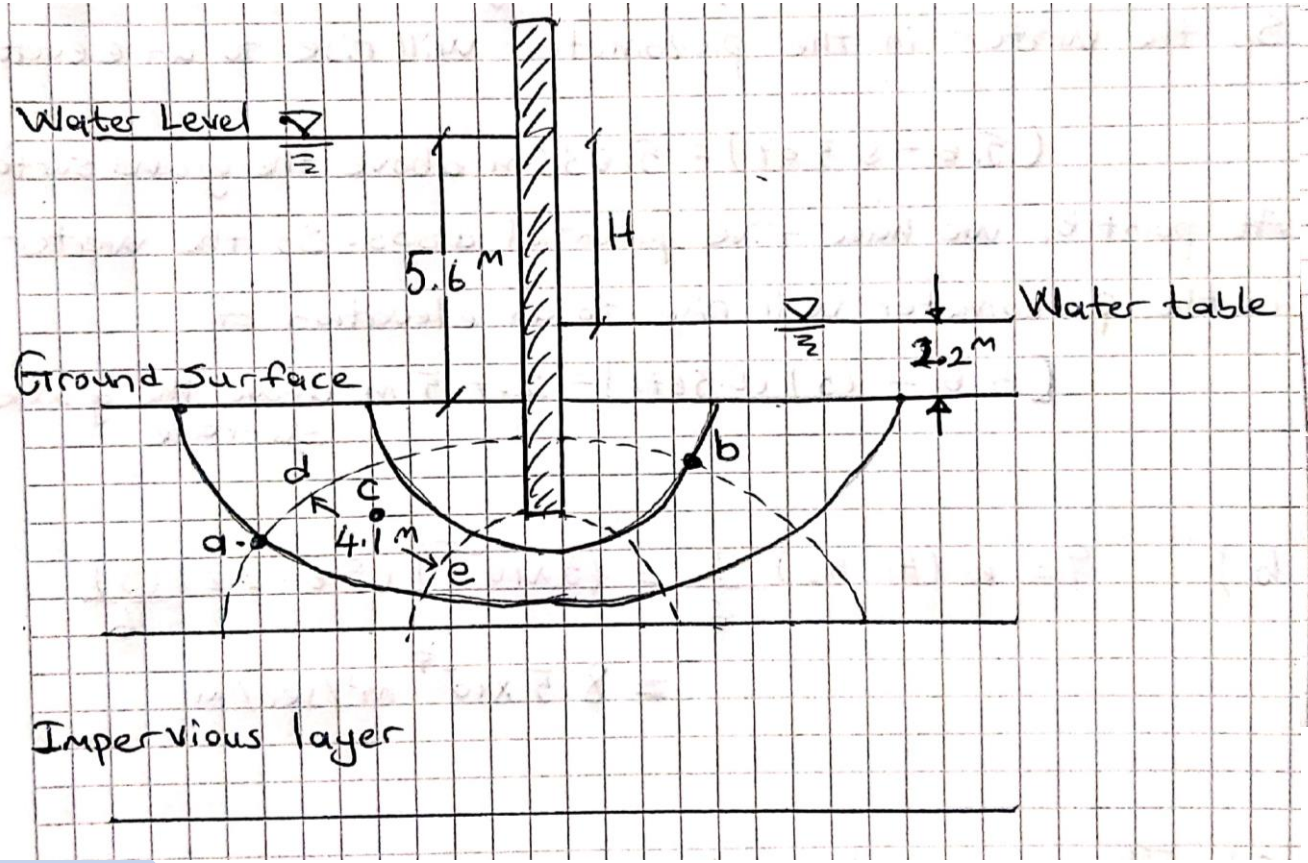
Example 3 (Optional)



Estimate the ratio of equivalent hydraulic conductivity

$$\frac{k_{H(\text{eq})}}{k_{V(\text{eq})}} = 145.4$$

Example 4



A flow net for flow around a single row of sheet piles in a permeable soil layer is shown in the figure. Given that $k_x = k_z = k = 5 \times 10^{-3}$ cm/sec, determine

a) How high (above the ground surface) the water will rise if piezometers are placed at points a and b

b) The total rate of seepage through the permeable layer per unit length

c) The approximate average hydraulic gradient at c

Solution 4

From the figure, we have $N_d = 6$, $H_1 = 5.6\ m$, and $H_2 = 2.2\ m$

So the head loss of each potential drop is

$$\Delta H = \frac{H_1 - H_2}{N_d} = \frac{5.6 - 2.2}{6} = 0.567\ m$$

At point a, we have gone through one potential drop.

So the water in the piezometer will rise to an elevation of

$(5.6 - 0.567) = 5.033\ m$ above the ground surface

At point b, we have five potential drops. So the water in the piezometer will rise to an elevation of

$[5.6 - (5)(0.567)] = 2.765\ m$ above the ground surface

Solution 4

b) $$q = k(H_1 - H_2)\frac{3}{6} = (5 \times 10^{-5})(5.6 - 2.2)\frac{(3)}{6}$$

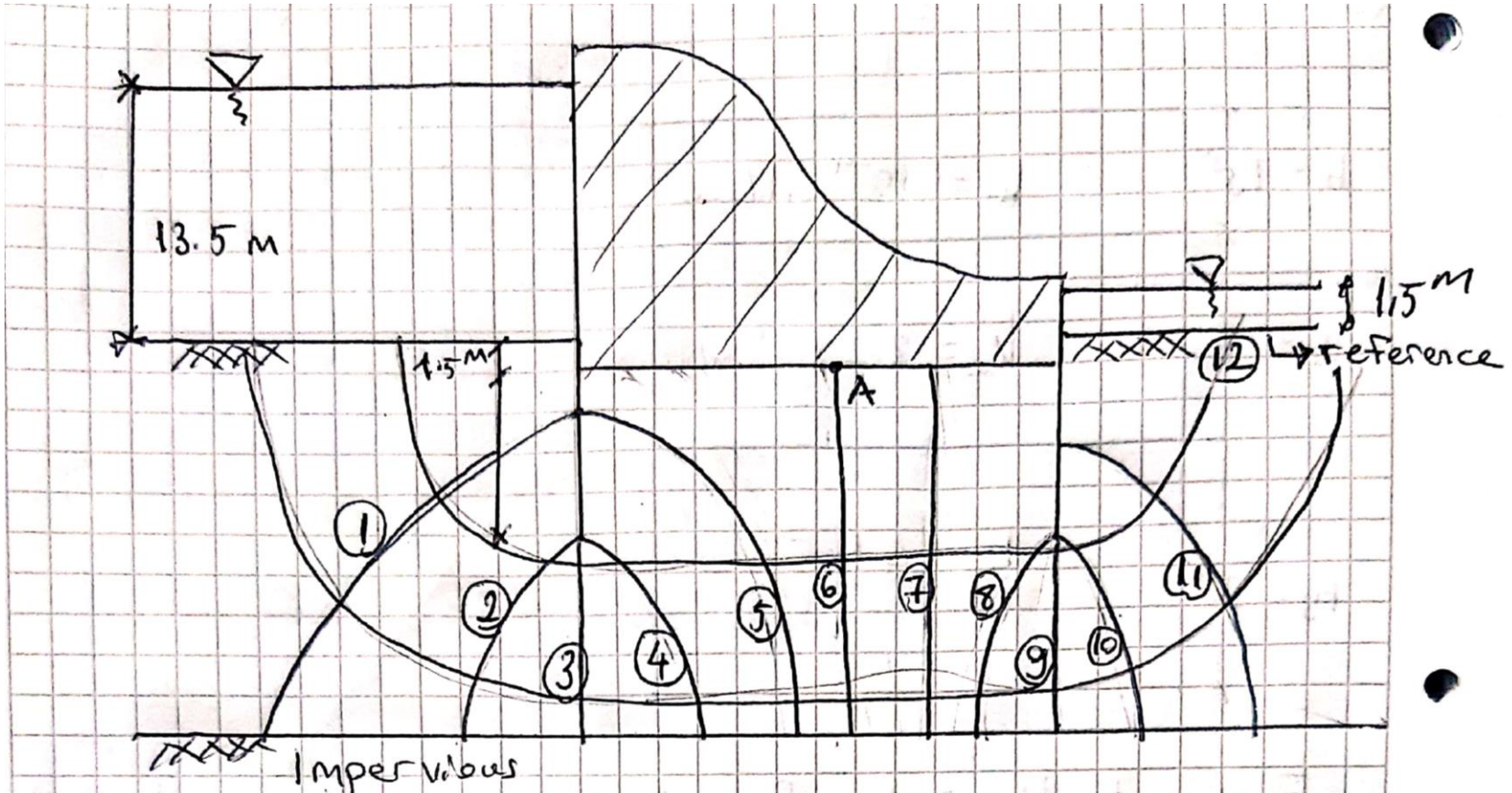
$$= 8.5 \times 10^{-5}\ m^3/sec/m$$

c) The average hydraulic gradient at c can be given as

$$i = \frac{\text{head loss}}{\text{average length of flow between d and e}}$$

$$= \frac{\Delta H}{\Delta L} = \frac{0.567\ m}{4.1\ m} = 0.138$$

Example 5



A dam section is shown in ~~the~~ the figure. The hydraulic conductivity of permeable layer is 1.3×10^{-5} m/sec $\gamma_{sat} = 19$ kN/m^3

a) Calculate the seepage loss under dam

b) calculate the water pressure at point A.

Solution 5

a) $q = k \frac{H N_f}{N_d}$

N_f: 3 (how many flow channel)

N_d: 12 (how many potential drops)

$H = 13.5 - 1.5 = 12^M$
(Head difference between the upstream and downstream sides)

$$q = 1.3 \times 10^{-5} \times 12 \times \frac{3}{12} = 3.9 \times 10^{-5} \underbrace{m^3/sec/m}_{1^M \text{ through figure plane}}$$

Solution 5

b) $h = \frac{u}{\gamma_w} + z$

$h - z = \frac{u}{\gamma_w} \Rightarrow u_A = \gamma_w (h_A - z_A)$ → below reference level (-1.5^M)

$u_A = \gamma_w (h_A + z_A)$

$h_A = H - 6.\Delta H$, $\Delta H = \frac{H}{N_d}$

$h_A = 12 - 6 \cdot \frac{12}{12} = 6^M$

$u_A = 9.81 (6 + 1.5) = 73.58 \ kN/m^2$

water pressure at point A.